

ViRel-Budget: Reliability-Constrained Visual-Token Budgeting for Green VLM Inference

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Sean Wan, Shilin Ou, and Luyao Zhang

Duke Kunshan University

Token sparsity alone does not establish reliable or green VLM inference.

Why current evaluation is insufficient

- Average accuracy can conceal weakened dependence on visual evidence.
- Pruning overhead can erase latency or device-energy savings.
- One favorable run cannot support a reproducible green claim.

Research questions

1. Can a frozen controller satisfy a prospective unsafe-acceptance limit?
2. How do trade-offs vary by model scale and pruning method?
3. Which qualified configurations reproduce measured energy savings?

Context: FastV and SCOPE [1, 2]; Green AI reporting [3, 4].

01

Strict evidence definition. Intervention-defined labels distinguish answer correctness from preservation of dense visual behavior.

02

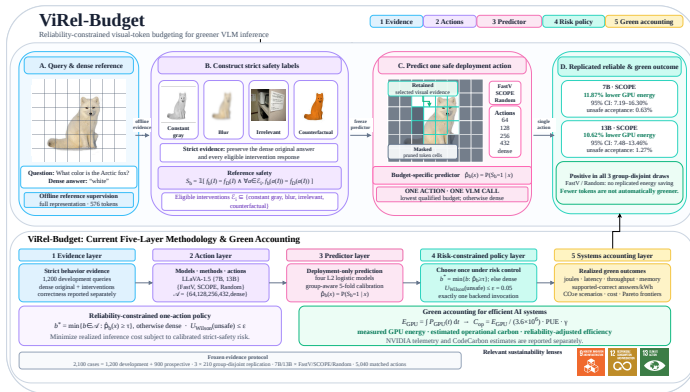
Frozen deployment policy. Budget-specific predictors select 64, 128, 256, 432, or dense inference without VLM fine-tuning or deployment-time labels.

03

Measured green validation. Paired GPU telemetry and three independent draws determine whether token sparsity yields device-level energy savings.

Deployment contract. One action is selected before inference. Controller overhead, dense fallback, and every unsafe acceptance remain inside the evaluation boundary.

ViRel-Budget Framework



Offline interventions define strict safety. A frozen predictor then chooses one deployment action under risk control.

Strict Safety and Frozen Policy



Original



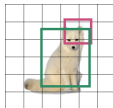
Counterfactual



Blur



Gray



Retained tokens

Strict safety label

$$S_{i,m,b} = A_{i,m,b} \wedge R_{i,m,b}$$

An action must preserve the dense original answer and every eligible intervention response. Task correctness remains separate.

Decision rule. Choose the smallest budget whose predicted safety exceeds its calibrated threshold; otherwise use dense inference.

$\leq 5\%$

Predeclared unsafe-acceptance limit

1

VLM call per query

Dense

Fallback when no pruned action
qualifies

Component	Design
Datasets	POPE, MMStar, Visual CounterFact [5–7]
Development	1,200 queries for labels, predictors, and frozen calibration
Prospective Replication	900 group-isolated queries for one-action reliability Three independent group-disjoint draws of 210 queries
Backends	LLaVA-1.5 7B and 13B
Pruning	FastV, SCOPE, deterministic Random
Actions	64, 128, 256, 432 visual tokens, or dense

Protocol safeguards

- Group-aware five-fold cross-validation.
- Controller frozen before prospective evaluation.
- Matched Dense references and counterbalanced order.
- Labels revealed after decisions are recorded.

Prospective reliability and energy replication use distinct held-out populations.

Prospective Reliability and Coverage

6/6

controllers satisfy the 5%
risk gate

1.67%

unsafe acceptance, 7B
SCOPE

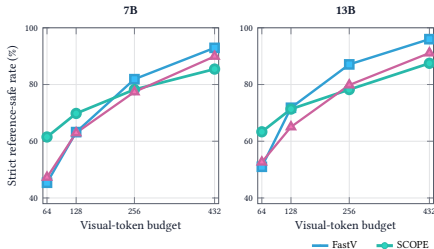
2.89%

unsafe acceptance, 13B
SCOPE

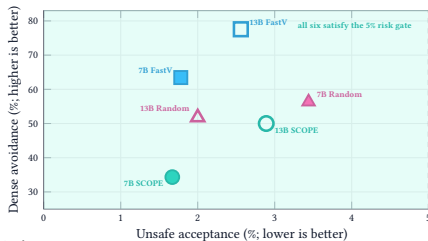
34.33–77.56%

reliability-qualified dense
avoidance

(a) Budget-specific strict safety



(b) Prospective reliability-coverage frontier



Safety generally improves with budget, but method rankings change across budgets and model scales.

Replicated Green Systems Outcome

SCOPE is the only evaluated pruning substrate that passes Gate R at both model scales.

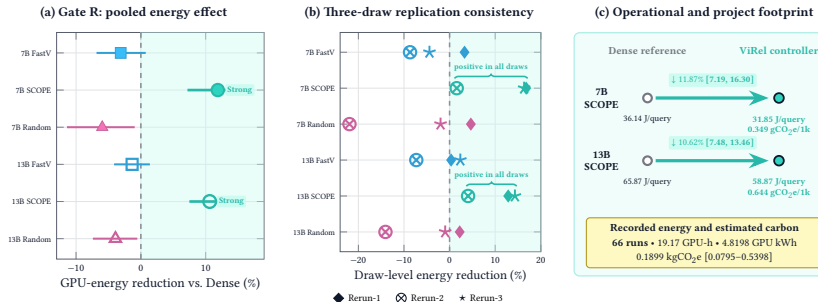


Figure 4: SCOPE consistently saves energy at both scales; project-level cost is reported separately.

Gate R requires acceptable risk, lower energy in every draw, and a pooled 95% confidence interval above zero.

Token Savings and Energy Savings Diverge

Scale	Method	Unsafe (%)	Dense avoid (%)	ΔE (% , $\uparrow$)
7B	FastV	1.78	63.44	-3.11
7B	SCOPE	1.67	34.33	+11.87
7B	Random	3.44	56.33	-5.96
13B	FastV	2.56	77.56	-1.35
13B	SCOPE	2.89	50.00	+10.62
13B	Random	2.00	51.78	-3.92

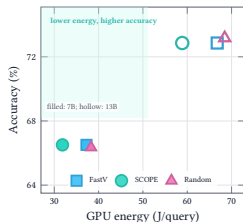
Observed pattern

- FastV achieves the highest dense avoidance, yet its pooled energy intervals include zero.
- Random removes tokens but uses more energy than its matched Dense reference.
- SCOPE couples acceptable risk with replicated energy savings.

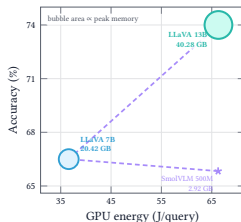
Dense avoidance measures algorithmic sparsity. Device energy determines whether sparsity becomes a green outcome.

Reliability-Qualified Pareto Frontiers

(a) Controller energy-accuracy frontier



(b) Dense model-scale Pareto frontier



(c) ViRel/SCOPE compute profile

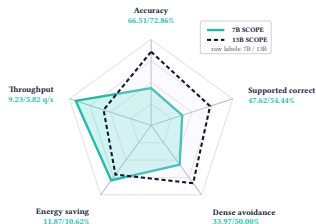


Figure 5: SCOPE defines the reliable energy frontier, while model scale governs the accuracy-memory-throughput trade-off.

7B SCOPE

31.85 J/query; 9.23 queries/s

7B favors energy productivity and throughput; 13B favors accuracy and supported-correct performance.

13B SCOPE

58.87 J/query; 5.82 queries/s

Reproducible System and Green Accounting

Repository architecture

- `pipeline.py` Dense references, interventions, and budget evaluations.
- `policies.py` Calibration, action selection, and dense fallback.
- `metrics.py` Reliability and resource-normalized metrics.
- `reporting.py` Bootstrap intervals and Pareto summaries.

```
bash scripts/run_virel_pipeline.sh scope  
paper1200
```

Measurement boundary

- NVIDIA power telemetry is sampled every 200 ms.
- CodeCarbon 3.2.9 provides run-level emissions context [8].
- Measured GPU energy remains separate from estimated carbon.

Ledger	Recorded project footprint
Runs	66
GPU time	19.17 GPU-h
GPU energy	4.8198 kWh
Central estimate	0.1899 kgCO ₂ e

Limitations and Claim Boundary

Behavioral evidence

Intervention-defined safety measures fidelity to dense behavior. It does not certify semantic grounding or causal use of the correct image region.

Predictor transfer

Task and source indicators are strong features. Their value may weaken on unseen task families or deployment shifts.

Systems dependence

Energy outcomes depend on the implementation, accelerator, batching, kernels, and execution order.

Backbone scope

Evidence covers LLaVA-1.5 7B and 13B with FastV, SCOPE, and Random. Cross-backbone transfer remains untested.

Note: The reported risk control is empirical under the evaluated distribution, not a distribution-free guarantee.

Green VLM inference requires reliability-qualified and directly measured systems savings.

01

Reliability. All six controllers satisfy the prospective 5% unsafe-acceptance limit.

02

Energy. SCOPE reduces GPU energy by 11.87% at 7B and 10.62% at 13B across three independent draws.

03

Interpretation. FastV and Random confirm that token reduction alone does not establish an energy benefit.

Open code

<https://github.com/StableTradeAtlas/ViRel-Budget>

Open data

<https://huggingface.co/datasets/StableTradeAtlas/ViRel-Budget>

Thank you

Questions and discussion

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